



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VALLI DEVAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 143.25 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.37305, 78.156916)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 95F4+6Q Bannihalli involves travel through several towns and larger areas such as Kangeyampalayam, Padaiveedu, Salem, Tharamangalam, Omalur, Thoppur, and Seshampatti. The journey spans approximately 143.25 kilometers, primarily using major highways and

connecting roads. The route provides access to commercial and rural areas, requiring the navigation of diverse driving environments.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate with dry winters, hot summers, and monsoons from June to September. Rain during monsoon can lead to waterlogged roads and reduced visibility. Drivers should be cautious of these conditions, especially in low-lying areas prone to flooding.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak traffic hours coincide with the morning (7:30 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) commuter rush. Areas around Salem and other major urban points tend to be more congested.
- **Congestion-Prone Areas:** Salem, being a commercial hub, can experience significant traffic congestion. Construction or roadworks near Thoppur could also cause potential delays.

4. Assessment of Road Quality and Infrastructure

- The roads vary from well-maintained highways to narrower, less maintained connecting roads. The SH 86 and Bhavani Main Road are generally in good condition but can have potholes or uneven surfaces in rural stretches.
- Road markings and signage might be less frequent or visible in rural areas, requiring additional caution.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route 1:** In severe congestion or roadblock scenarios in Salem, consider detouring via local roads parallel to the highway.
- **Alternative Route 2:** Utilizing state highways alongside NH44 near Thoppur for smoother transit, depending on real-time traffic data.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu regulations require permits for transporting hazardous materials, compliance with safety standards, and clearly displayed hazard placards. Time restrictions on heavy vehicles may vary, requiring attention to local signs or enforcement checkpoints.

7. Overview of Historical Incidents

- Limited data on historical incidents specific to hazardous materials in this exact route. However, Salem has witnessed a few heavy vehicle accidents due to speeding and improper vehicle maintenance, particularly in traffic-dense areas.

- Awareness of past incidents encourages vigilance regarding vehicle maintenance and adherence to speed limits.

8. Environmental Considerations and Sensitive Areas

- Rural areas en route are ecologically sensitive, particularly around agricultural land. Efforts should be made to minimize environmental impact through controlled driving (avoiding excessive idling and emissions).
- Watch for regional wildlife crossings in rural stretches, especially during dusk or dawn.

9. Analysis of Communication Coverage

- Generally, communication coverage is good along the major segments, such as highways.
- Potential dead zones are more likely in rural and remote stretches, especially around forested or hilly terrains.

10. Estimated Emergency Response Times

- **Salem and Thoppur:** As these are larger urban centers, emergency response times may be as quick as 15-30 minutes.
- **Rural segments:** Emergency services could take up to 45-60 minutes or more, depending on road conditions and traffic.

11. An Overall Summary of Risk Assessment

This route encompasses major highways with good infrastructure and certain rural roads requiring caution. The primary risks include weather-related conditions during monsoons, potential traffic congestion, and road quality challenges in less urbanized stretches. Regular communication checks and compliance with local regulations are critical for safe transit. Emergency preparedness measures should be a priority due to variable response times.

In summary, while this route is relatively straightforward, vigilance and preparation for variable traffic and weather conditions, along with observance of safety protocols, will be essential for ensuring safe passage of heavy vehicles carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4749755, 77.869149	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44028, 77.87541	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	Medium	11.46392, 77.87328	30 KM/Hr
6	Turn	High	11.47391, 77.86844	15 KM/Hr
7	Blind Spot	Blind Spot	11.47481, 77.86920	10 KM/Hr
8	Turn	Medium	11.47354, 77.86775	30 KM/Hr
9	Turn	Medium	11.45543, 77.81780	30 KM/Hr
10	Turn	Medium	11.45580, 77.81697	30 KM/Hr
11	Turn	High	11.45615, 77.81677	15 KM/Hr
12	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
13	Turn	Medium	11.49377, 77.88573	30 KM/Hr
14	Turn	High	11.49469, 77.88488	15 KM/Hr
15	Turn	High	11.55618, 77.89789	15 KM/Hr
16	Turn	Medium	11.55629, 77.89888	30 KM/Hr
17	Turn	High	11.58228, 77.90841	15 KM/Hr
18	Turn	High	11.58340, 77.90637	15 KM/Hr
19	Turn	Medium	11.59201, 77.90649	30 KM/Hr
20	Turn	Medium	11.63882, 77.94859	30 KM/Hr
21	Turn	Medium	11.64161, 77.95177	30 KM/Hr
22	Turn	Medium	11.64226, 77.95200	30 KM/Hr
23	Turn	Medium	11.64716, 77.95118	30 KM/Hr
24	Turn	Medium	11.69553, 77.97110	30 KM/Hr
25	Turn	Medium	11.69806, 77.97537	30 KM/Hr
26	Turn	Medium	11.71386, 77.99552	30 KM/Hr
27	Turn	Medium	11.71735, 78.00019	30 KM/Hr
28	Turn	Medium	11.71915, 78.01106	30 KM/Hr
29	Turn	Medium	11.72492, 78.01660	30 KM/Hr
30	Turn	Medium	11.72765, 78.02291	30 KM/Hr
31	Turn	Medium	11.73786, 78.03505	30 KM/Hr
32	Turn	High	11.73652, 78.03866	15 KM/Hr
33	Turn	Medium	11.74255, 78.04707	30 KM/Hr
34	Turn	Medium	11.77984, 78.05782	30 KM/Hr
35	Turn	Medium	11.86902, 78.08380	30 KM/Hr
36	Turn	Medium	11.87988, 78.08176	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
37	Turn	Medium	11.92637, 78.06515	30 KM/Hr
38	Turn	Medium	11.93726, 78.05359	30 KM/Hr
39	Turn	Medium	11.95482, 78.04687	30 KM/Hr
40	Turn	High	11.95837, 78.04986	15 KM/Hr
41	Turn	High	11.96007, 78.04945	15 KM/Hr
42	Turn	Medium	11.96347, 78.05142	30 KM/Hr
43	Turn	High	11.96415, 78.05126	15 KM/Hr
44	Turn	Medium	11.96519, 78.04958	30 KM/Hr
45	Turn	High	11.96620, 78.04920	15 KM/Hr
46	Turn	Medium	11.96842, 78.05097	30 KM/Hr
47	Turn	Medium	12.12010, 78.13282	30 KM/Hr
48	Turn	Medium	12.26535, 78.08429	30 KM/Hr
49	Turn	Medium	12.27235, 78.08086	30 KM/Hr
50	Turn	High	12.29602, 78.07393	15 KM/Hr
51	Turn	High	12.29655, 78.07551	15 KM/Hr
52	Turn	High	12.31207, 78.07838	15 KM/Hr
53	Turn	High	12.31355, 78.07485	15 KM/Hr
54	Turn	Medium	12.31427, 78.07473	30 KM/Hr
55	Turn	High	12.32003, 78.07652	15 KM/Hr
56	Turn	Medium	12.32051, 78.08341	30 KM/Hr
57	Turn	High	12.32340, 78.08687	15 KM/Hr
58	Turn	Medium	12.32338, 78.09546	30 KM/Hr
59	Turn	Medium	12.33674, 78.10807	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Ck Hospital	11.473504, 77.8650523	30 km/h	Medium
7	hospital	SP Hospital	11.4708567, 77.8654872	30 km/h	Medium
8	hospital	Lakshmi Hospital	11.4723746, 77.8646891	30 km/h	Medium
9	hospital	ESI Hospital	11.4717067, 77.8642668	30 km/h	Medium
10	clinic	Sutha Clinic	11.473718, 77.862614	30 km/h	Medium
12	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
13	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
14	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
15	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
16	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
17	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
18	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
19	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
20	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
21	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
22	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
24	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
25	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
26	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
27	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
28	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
29	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
30	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
31	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
32	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
33	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
34	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
38	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
40	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
41	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
42	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
43	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
11	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
37	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
39	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4749755, 77.869149



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46392, 77.87328



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47391, 77.86844



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.47481, 77.86920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47354, 77.86775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81780



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45580, 77.81697



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59201, 77.90649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63882, 77.94859



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64161, 77.95177



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64226, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64716, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69553, 77.97110



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69806, 77.97537



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71915, 78.01106



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72492, 78.01660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73786, 78.03505

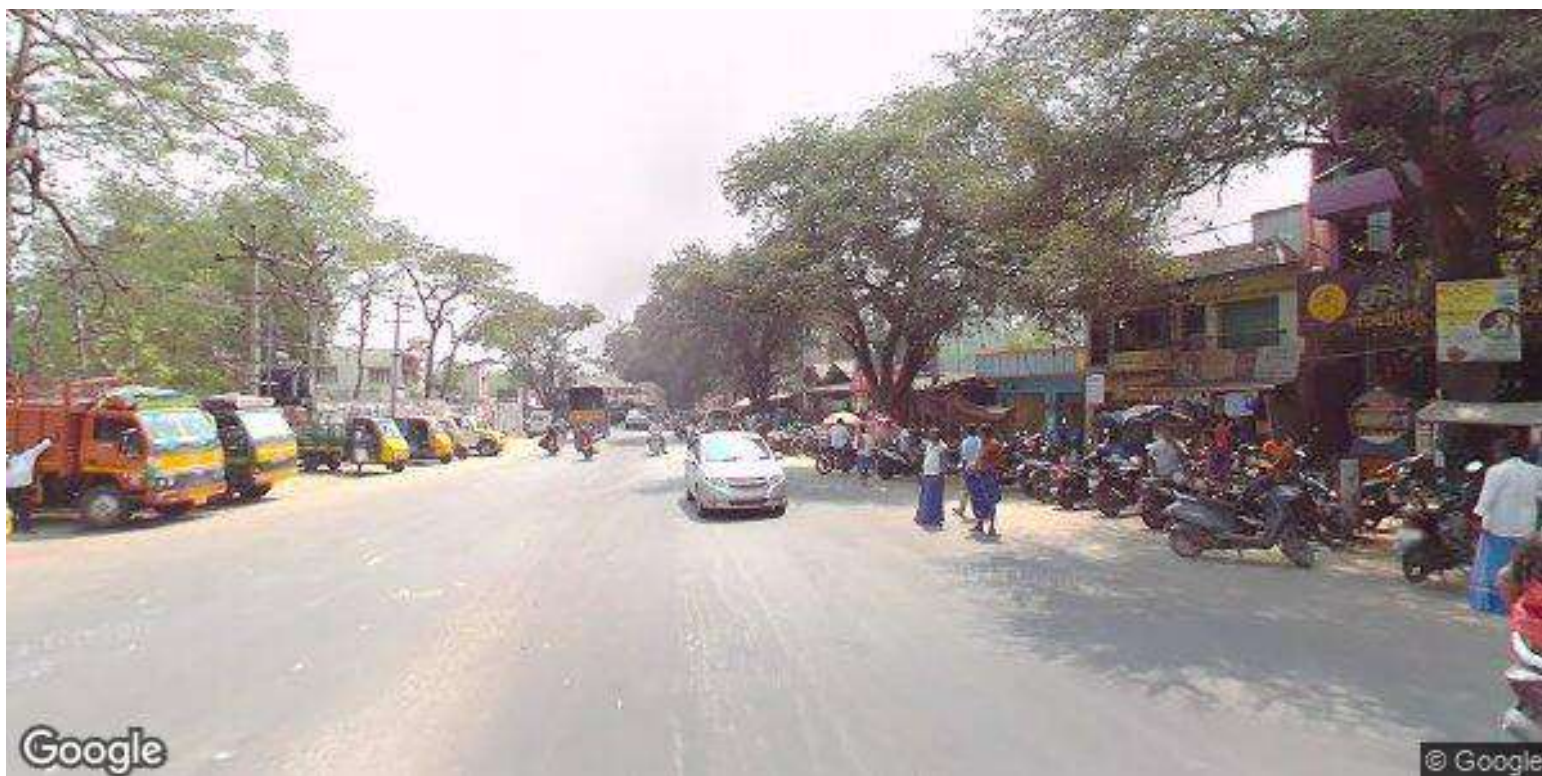


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74255, 78.04707



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77984, 78.05782



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86902, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87988, 78.08176



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92637, 78.06515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93726, 78.05359



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95482, 78.04687



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95837, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96007, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96347, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96415, 78.05126



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96519, 78.04958



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96620, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96842, 78.05097



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26535, 78.08429



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.27235, 78.08086



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29602, 78.07393



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29655, 78.07551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31207, 78.07838



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31355, 78.07485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31427, 78.07473



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.32003, 78.07652



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32051, 78.08341



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.32340, 78.08687



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32338, 78.09546



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.33674, 78.10807
